

The Impact of Dark Matter Mass Fraction on the Type of Spiral Structure in Disc Galaxies

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ABSTRACT

This paper explores the relationship between the presence and details of spiral structure and the baryonic mass fraction of disk galaxies, making use of both spiral arm classifications from Galaxy Zoo DESI and HI emission line data from the ALFALFA-SDSS Galaxy Catalog. The study focuses on observable differences in the trends in three arm classes — grand design, flocculent, and multiple arm — using unsupervised machine learning approaches (namely K-means clustering and Principal Component Analysis, PCA) to generate these classifications from Galaxy Zoo votes of the numbers of visible spiral arms. Our results suggest a correlation between higher baryonic mass fraction in disk galaxies and the presence of clearly visible spiral structure. We interpret this as heavier discs being more prone to developing spiral instabilities. This study contributes to our understanding of the connection between spiral structure, disk mass and dark matter halos, and also demonstrates the utility of combining human visual classifications with machine learning techniques to obtain the best possible classifications.

Key words: Spiral galaxy — Baryonic mass fraction — Galaxy morphology — Unsupervised machine learning — Arm class classification

1 INTRODUCTION

Understanding the structure and evolution of galaxies is a cornerstone of modern astrophysics. Among the diverse galaxy morphologies seen in our Universe (e.g. [de Vaucouleurs et al. 1991](#); [Willett et al. 2013](#)), spiral galaxies stand out due to their beauty and complexity. However, the spiral structures seen in galaxies are not merely aesthetic; they provide valuable insights into the physical processes governing galaxy formation and evolution (e.g. [Dobbs & Baba 2014](#); [Sellwood & Masters 2022](#)).

Spiral patterns, shaped by the interplay of gravitational forces, star formation, and gas dynamics, are dynamical features in disk galaxies, believed to be influenced by both internal factors (such as galaxy mass, gas content, bar properties and more, e.g. [Hart et al. 2018](#); [Díaz-García et al. 2019](#); [Smith et al. 2024](#)) including the surrounding environment of the galaxy ([Elmegreen et al. 1982](#); [Ann 2014](#)). One key aspect of studying spiral galaxies is their classification into three distinct arm classes: “grand design,” “multiple arm,” and “flocculent” galaxies first introduced by [Elmegreen & Elmegreen \(1984, hereafter E84\)](#). In the language of E84, grand design galaxies are characterized by highly symmetric and continuous spiral arms, prototypically in a two-arm configurations. Flocculent galaxies display fragmented and patchy arms, often appearing as “many-armed” structures. Finally, multiple arm galaxies exhibit several discernible arms, and suggested by E84 as an intermediate morphological class between grand design and flocculent galaxies. These spiral arm type classifications likely reflect underlying physical mechanisms, such as the strength of the density waves and/or local gravitational instabilities, that drive spiral

arm formation ([Elmegreen & Elmegreen 1987a](#); [Dobbs & Baba 2014](#); [Sellwood & Masters 2022](#)).

Grand design galaxies, with their large scale bi-symmetric arms, are usually assumed to have spiral arms which are density waves, where the spirals act as wave-like patterns propagating through the galactic disk. These density waves may be tidally generated (e.g. [Dobbs et al. 2010](#)), driven by galactic bars ([Athanasoula 2012, e.g.](#)), or represent a self-excited superposition of modes can have the same effect (e.g. [Sellwood & Carlberg 2014](#)). Meanwhile, the short, fragmented and very numerous arms of flocculent galaxies are believed to arise from localized gravitational instabilities (e.g. [Elmegreen & Thomasson 1993](#)), and or turbulence in the interstellar medium (ISM, [Elmegreen et al. 2003](#)).

The role of dark matter halos in shaping the gravitational potential within galaxies underscores their influence on spiral structures. Observational studies, such as [Hart et al. \(2018\)](#), highlight the significant role of dark matter halos in shaping the gravitational potential, which influences the dynamics and morphology of spiral structures within galaxies.

The environment surrounding galaxies also plays a pivotal role in their cosmic evolution, including their morphology (e.g. the classic density-morphology relation; [Dressler 1980](#)). Galaxy density, for example as defined as the projected number of neighboring galaxies within a given area (e.g., Mpc^2), provides a measure of local galaxy clustering. It has been observed that high-density environments often host red sequence and/or elliptical galaxies, while lower-density regions favor blue sequence, spiral galaxies (e.g. [Lee et al. 2010](#)). This bimodal distribution highlights the interplay between stellar mass, environmental density, and galaxy morphology, suggesting that local density significantly impacts star formation and spiral structure stability. Indeed, [Elmegreen \(1980\)](#) analyzed spiral arm spurs and

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proposed that their formation and stability are likely influenced by density wave processes rather than purely local or stochastic events, indicating a substantial role of large-scale environmental factors on spiral morphology and associated star formation features.

In galaxy astronomy, the "environment" refers broadly to both local and global conditions, including galaxy density and position within larger cosmic structures such as clusters and superclusters, which significantly influence galaxy morphology and evolution (Tempel et al. 2011). The spiral arms of galaxies, specifically, have been suggested to exhibit dependencies on these environmental factors and the associated dark matter halo properties. Observational studies have demonstrated relationships between spiral arm morphology, measured as pitch angle, and the galaxy rotation curve characteristics, which directly reflect the mass distribution within galactic halos (Seigar et al. 2006). For instance, the maximum rotational velocity of spiral galaxies has been empirically linked to spiral-arm pitch angles, suggesting tighter spiral arms correspond to galaxies with higher rotational velocities and therefore more massive dark matter halos (Davis et al. 2019).

In this study, we explore the relationship between spiral arm properties and baryonic mass fractions in galaxies using a sample of galaxies with morphologies from Galaxy Zoo (GZ) analysis of images in the Dark Energy Spectroscopic Instrument (DESI) survey (Walmsley et al. 2023; Dey et al. 2019), and HI rotation widths from the The Arecibo Legacy Fast ALFA (ALFALFA) survey (Haynes et al. 2018). By exploring these correlations across different arm classes, we aim to advance our understanding of how internal and environmental factors shape the formation and evolution of spiral structures in disk galaxies.

We introduce the data and sample selection in §2, detailing the criteria and methodologies employed to curate our galaxy sample. In §3, we classify spiral structures into grand design, multiple arm, and flocculent galaxies using Galaxy Zoo DESI alongside unsupervised machine learning techniques. The results of our analysis are presented in §4, where we explore the dependence of arm class on baryonic and dynamical masses (§4.1) and gas content (§4.2). Additionally, we examine the role of galaxy environment in determining spiral structures in §4.3.

2 DATA AND SAMPLE SELECTION

2.1 Galaxy Morphology Data

2.1.1 Galaxy Zoo: DESI

We start our sample selection using the galaxy morphology dataset "Galaxy Zoo DESI" (Walmsley et al. 2023) which are based on Galaxy Zoo analysis of the DESI Legacy Survey (Dey et al. 2019) DR8 images. Galaxy Zoo (GZ) involves citizen scientists examining galaxy images to gather morphology data. In GZ, many tens of citizen scientists review each galaxy image and make choices from provided options to classify them. Galaxy morphology data is thus expressed as numbers indicating the fraction of citizen scientists who observed specific features in each galaxy. The GZ DESI data release (Walmsley et al. 2023) makes use of these volunteers' classification "vote fractions" for a subset of the sample to train the Zoobot deep learning algorithm to predict volunteer classifications for the entire DESI DR8 samples (Walmsley et al. 2023), and these are the classifications we make use of in this work to identify spiral features.

We make use of various questions related to spiral structure in GZ:DESI. The part of the GZ:DESI decision tree we make use of

is illustrated in Figure 1). Specifically we are interested in votes on the presence of spiral structure (which is asked after a person indicates the galaxy has features and is not an edge-on disc), and also classifications of the number of arms present, which is one of the questions asked after volunteers identify spiral structure.

2.1.2 Spiral Types from S4G

Galaxy Zoo does not directly ask questions about spiral type (i.e. asking for classifications of grand design, multi-arm or flocculent spirals). For these identifications, we make use of data from the Spitzer Survey of Stellar Structure in Galaxies (S4G; Muñoz-Mateos et al. 2013; Sheth 2012; Sheth et al. 2010; Watkins et al. 2022). This survey of 485 galaxies, includes a detailed morphological catalog compiled by visual inspection by professional astronomers and including spiral type (Elmegreen et al. 2011; Buta et al. 2015). This dataset categorizes 398 spiral galaxies in S4G into three different spiral classes: grand design (GD), multiple arm (M), and flocculent (F).

We cross match the S4G spiral sample with GZ:DESI and ALFALFA data; this results in a sample of 341 spiral galaxies with both S4G spiral type classifications and the GZ:DESI morphologies. Hereafter we call this our S4G subset.

Cold Gas Data from ALFALFA

We incorporate neutral-hydrogen measurements and derived dynamical masses from the Arecibo Legacy FastALFA (ALFALFA) Survey (Haynes et al. 2018). ALFALFA was an untargeted 21cm drift-scan survey carried out with the seven-feed Arecibo L-band Feed Array (ALFA) between 2005 and 2011, covering $\approx 7000, \text{deg}^2$ of high-Galactic-latitude sky at a typical rms sensitivity of $\sim 2.5, \text{mJy}$ per 10km s^{-1} channel (see Giovanelli et al. 2005, for full technical details). We make use of the Yu et al. (2022, hereafter Y22), analysis of ALFALFA, which provided optimal SDSS matches, parameters and re-measurements of ALFALFA HI line parameters including dynamical masses.

To build a homogeneous set of optical morphologies, stellar masses and H I properties, we performed a positional cross-match of two publicly-available value-added tables using TOPCATv4.8:¹

(i) GZDESI's "external_catalog"², a cross match of the GZ DESI morphology catalog with other information;

(ii) Table 2 and Table 3 of Y22, providing ALFALFA based measurements.

2.2 Galaxy Masses

In this work, we are interested in assessing correlations between spiral features and the total, stellar, HI mass and baryonic mass fraction of galaxies.

Estimates of the stellar mass and dynamical mass are taken from the SDSS-ALFALFA-GSWLC catalog compiled by Yu et al. (2022), which provides values derived from SED fitting and rotation velocity modeling respectively.

The stellar masses in Yu et al. (2022) come from the GALEX-SDSS-WISE Legacy Catalog (GSWLC-X2; Salim et al. 2018).

¹ <http://www.star.bris.ac.uk/~mbt/topcat/>

² Hosted on Zenodo - see link at <https://data.galaxyzoo.org/#section-20>

The HI mass comes directly from ALFALFA HI flux measurements using the standard relation

$$M_{\text{HI}} = 2.36 \times 10^5 D_L^2 F_{\text{HI}}, \quad (1)$$

where F_{HI} is the integrated 21 cm line flux in Jy km s^{-1} and D_L is the luminosity distance in Mpc. This method follows Haynes et al. (2018) and Giovanelli et al. (2005).

We estimate the total baryonic mass (M_{bary}) by adding up the gaseous mass and stellar mass of the galaxies, using the approximation

$$M_{\text{bary}} = 1.07 M_{\star} + \frac{4}{3} M_{\text{HI}} \quad (2)$$

which (as explained by Goddy et al. 2023) accounts for both cosmic He ($M_{\text{gas}} = \frac{4}{3} M_{\text{HI}}$ considering the universe’s hydrogen composition of approximately 75% by mass) and includes a small correction for the contribution of molecular hydrogen to the total mass assuming it is on average 7% of the stellar mass. Baryonic mass fraction is then obtained by dividing the baryonic mass by the total, or dynamical mass.

2.3 Sample Selection and Spiral Arm Number

Because we wish to investigate spiral structure, our sample construction deliberately excludes galaxies in which spiral arms cannot be recognised. We employed the previously derived ALFALFA–Galaxy Zoo DESI cross-matched catalogue, which comprises 12 951 unique galaxies. Applying the inclination cut $b/a > 0.4$ together with a ‘features or disc’ vote fraction $p_{\text{features}} > 0.430$, limiting the redshift to $0 < z < 0.05$ to bypass the San Juan radar gap, and retaining only sources with reliable mass estimates ($\log M_{\text{HI}_1} > 0$, $\log M_{\text{HI}_2} > 0$, $\log M_{\star} > 0$) reduces the data set to a parent sample of 7 908 galaxies—61.0 per cent of the original cross-match. To ensure stellar mass completeness and enhance statistical leverage, we further require $\log M_{\star} > 9.4$, which yields a final working sample of 5749 gas-rich, spiral-arm-bearing discs with secure ALFALFA detections (Fig. 2).

Observations reveal that most of our galaxies have baryonic mass fractions in the range of 0.05–0.4 (see Figure 3 below). In order to enhance the statistical robustness of our investigation, this paper narrows the focus to this specific range. Which limits the sample further to 5393 gas rich featured galaxies.

Within the context of GZ DESI, we select spiral galaxies with a vote fraction for the presence of spiral arms, $p_{\text{spiral}} > 0.5$. This limits the sample to 5079 gas rich spirals. High probability spiral galaxies are identified with a stricter criterion of $p_{\text{spiral}} > 0.8$. The vote fraction of volunteers indicating they see spiral structures in galaxies is related to the likelihood that a disk galaxy has spiral features present, so high spiral vote fraction galaxies exhibit clearer and more prominent spiral structures compared to other spiral galaxies. We define the “strong spiral” fraction as the ratio between the number of high spiral probability galaxies and the total number of spiral galaxies. The distribution of baryonic mass fraction for all spirals, and high probability spirals, as well as how the ‘strong spiral’ fraction varies with baryonic mass fraction is shown in Figure 3. As is clear, spiral galaxies with larger baryonic mass fractions are more likely to have “strong” or high vote fraction spirals.

We classify the 5079 spiral galaxies by their arm numbers with GZ DESI vote fraction answers for three number of spirals arms, $p_{1\text{-arm}}$, $p_{2\text{-arms}}$, $p_{3\text{-arms}}$, $p_{4\text{-arms}}$, and $p_{\text{cannot-tell}}$ respectively. A galaxy is clearly categorized as having “N-arms” if the probability parameter for having that type exceeds $p_{\text{N-arms}} > 0.5$. This classification

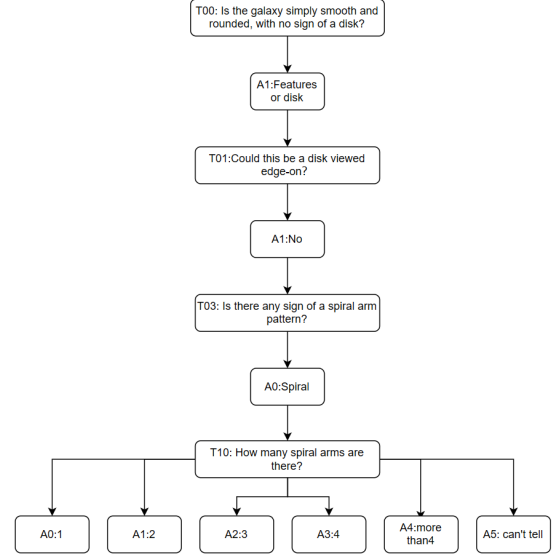


Figure 1. This figure illustrates the Galaxy Zoo morphology selection we use in this study, adapted from Hart et al. (2016), Figure 2. In this diagram, “T” numbers represents questions asked as part of the classification process, while “A” numbers denotes the different possible answer choices.

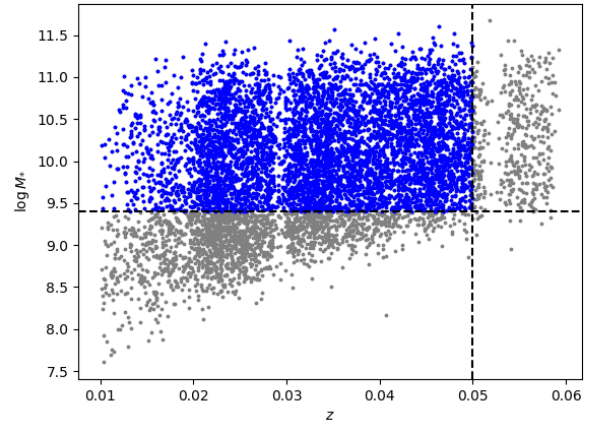


Figure 2. This plot shows the sample selection based on redshift and stellar mass. Each dot represents a galaxy in the sample, with blue dots indicating the selected galaxies.

methodology yields a classification of spiral galaxies clearly having each arm number, as presented in Table 1, but leaves the majority of the spirals in the sample with ambiguous arm counts. We consider that the galaxies have more than four arms and the galaxies that are classified as not to be determined are the many armed galaxies. We consider 2–arms and 4–armed galaxies as grand-design spiral galaxies, since they are bi-symmetric (the majority of these have 2-arms). Since this classification removes a large fraction of the spiral sample as “ambiguous” we also investigate making use of the full range of vote fractions for arm number below.

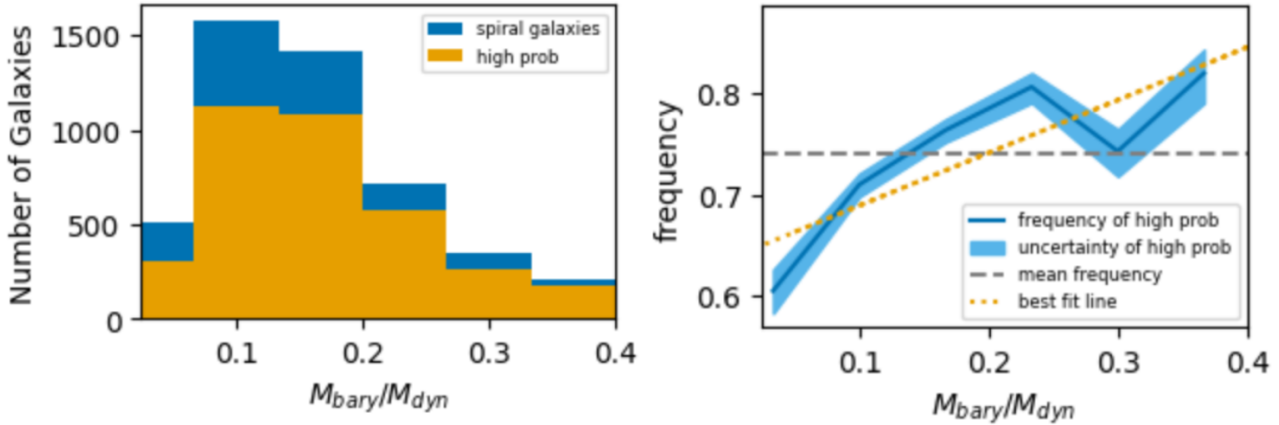


Figure 3. The two diagrams show the distribution of spiral galaxies based on their vote fraction of being classified as spiral ($p_{\text{spiral}} > 0.5$) and with even higher probabilities ($p_{\text{spiral}} > 0.8$). The frequency is expressed as a fraction: the numerator is the number of high-vote fraction spiral galaxies in each bin, and the denominator is the total number of spiral galaxies in the same bin. We focus on galaxies with a baryonic mass fraction in the range of 0.05–0.4, where most spiral galaxies lie. The left plot shows the count of galaxies across different baryonic mass fraction bins. The right plot is a frequency histogram illustrating the likelihood of galaxies being classified as high-probability spirals across these bins.

Table 1. Number of spirals in GZDES–ALFALFA sample with clearly identified different spiral arm numbers ($p_{\text{arm-count}} > 0.3$ and $p_{\text{other arm-count}} < 0.2$).

Spiral Galaxy Arm Count	Number of galaxies
1 arm	9
2&4 arms	842
3 arms	7
many arms	1404
ambiguous	3487

3 CLASSIFICATION AND ANALYSIS OF SPIRAL ARM STRUCTURES

Spiral arm structures offer key insights into disk galaxy formation and evolution. Based on appearance, spirals are classified as ‘grand design’ (symmetric, usually two-armed [Sellwood & Masters 2022](#)), ‘multiple arm’ (intermediate) and ‘flocculent’ (fragmented, many-armed) ([Elmegreen & Elmegreen 1984, 1987a](#)).

While the Galaxy Zoo survey did not explicitly ask participants to differentiate between flocculent and multiple arm galaxies, the resulting classifications of arm number may be possible to use to distinguish these two types. To test this, we employ unsupervised machine learning (ML) methods, specifically k-means clustering and Principal Component Analysis (PCA), to classify spiral arm configurations into distinct categories. We then compare these to expert classifications of spiral type from S4G.

3.1 Methodology for Classification

We first examined the effect of unsupervised ML algorithms on morphology classification using our sample with data in the S4G dataset ([Buta et al. 2015](#)), which we further limit to only spiral galaxies with a probability ($p_{\text{spiral}} > 0.5$) from GZ:DESI.

We employed Principal Component Analysis (PCA) in conjunction with the K-means clustering algorithm to analyze the data. PCA,

a linear dimensionality reduction technique, transforms correlated variables into a smaller set of uncorrelated variables known as principal components, effectively capturing the maximum variance within the dataset ([Jolliffe 2002](#)). This transformation facilitates the identification of underlying patterns by projecting the data onto a new coordinate system where the greatest variances by any projection of the data come to lie on the first coordinates (the principal components).

The K-means clustering algorithm aims to partition a dataset into K distinct, non-overlapping clusters by minimizing the within-cluster sum of squares ([MacQueen 1967](#)). The algorithm iteratively refines cluster assignments through the following steps:

- (i) **Initialization:** Select K initial centroids, which can be chosen randomly or by specific methods such as the Forgy or Random Partition methods ([Celebi et al. 2013](#)).
- (ii) **Assignment:** Assign each data point to the nearest centroid, forming K clusters based on the minimum Euclidean distance criterion.
- (iii) **Update:** Compute new centroids as the mean of the data points assigned to each cluster.
- (iv) **Iteration:** Repeat the assignment and update steps until convergence is achieved, indicated by no significant changes in cluster assignments or centroids.

This iterative process ensures that the total variance within each cluster is minimized, leading to more homogeneous groupings of data points. However, it is important to note that the K-means algorithm is sensitive to the initial placement of centroids and may converge to local optima; therefore, multiple runs with different initializations are often recommended to achieve robust results ([Arthur & Vassilvitskii 2007](#)).

In this work we performed a k-means clustering of a PCA transformation of spiral arm number classification vote fractions from GZ:DESI.

The first two principal components (PC1 and PC2), which capture

the majority of the morphological variance, are defined as

$$PC1 = -0.029p_{1\text{-arm}} + 0.824p_{2\text{-arms}} - 0.048p_{3\text{-arms}} - 0.109p_{4\text{-arms}} - 0.553p_{\text{cannot-tell}}, \quad (3)$$

and

$$PC2 = 0.032p_{1\text{-arm}} - 0.406p_{2\text{-arms}} + 0.343p_{3\text{-arms}} + 0.439p_{4\text{-arms}} - 0.723p_{\text{cannot-tell}}. \quad (4)$$

We compare the clustering of these PCA variables with manual classification of spiral type in Figure 5, which depicts the outcomes of a K-means clustering algorithm applied to categorize galaxies into three arm classes denoted as F (flocculent), G (grand design), and M (multiple arm) based on the S4G classifications. Evaluating the effectiveness of the K-means algorithm involves observing the consistency of color-coded points within each cluster. The plots reveal a certain level of alignment between manual classification and k-means clustering, particularly where clusters converge at specific angles (indicating small PC1 and small PC2, small PC1 and large PC2, large PC1). This alignment is most pronounced in regions that exhibit the best agreement with manual classification results. However, in the central areas of the plots, the mixed arrangement suggests that the algorithm's grouping does not perfectly align with astronomers' classifications. Consequently, the results indicate that k-means clustering in PCA space accurately classifies arm classes at extreme values of the PCA vectors, but exhibits limitations in the central regions.

While the standard morphological definition of grand design spirals is typically restricted to two-arm structures (Elmegreen & Elmegreen 1984), in this work we chose to include both two-arm and four-arm galaxies in the grand design category. This decision was driven by two considerations. First, four-arm galaxies constitute only a small fraction of the total sample only 18 4 arms galaxies exit and thus have limited impact on the overall cluster distribution. Second, the PCA loadings confirm that the four-arm vote fraction contributes minimally to the principal components. Specifically, in PC1 - which is most strongly aligned with the two-arm vote (coefficient of 0.824)—the coefficient for the four-arm vote is only -0.109. This suggests that the inclusion of four-arm galaxies does not significantly distort the morphological variance captured by the principal components. Therefore, merging the two- and four-arm categories for the purposes of grand design classification simplifies the clustering without compromising the classification fidelity.

To quantitatively evaluate the performance of the PCA+K-means classification, we constructed a confusion matrix comparing our cluster-based classifications to the manual S4G arm type labels. The matrix below shows the number of galaxies in each predicted cluster versus their true class as figure 4 shows.

This confusion matrix reveals that the algorithm performs reasonably well at identifying clear cases of grand design and flocculent galaxies, but struggles more with multiple arm types, which tend to be confused with both other categories. The relatively high number of misclassifications between multiple arm and flocculent galaxies suggests that their morphological features may be less distinct in PCA space, particularly within the interarm.

Using these methods, we classify spiral galaxies into three groups in the (PC_1, PC_2) plane with two linear decision boundaries derived from the five arm-number vote fractions:

$$y_1(PC_1) = 4PC_1 - 0.5, \quad y_2(PC_1) = -0.433PC_1 - 0.05,$$

which intersect at $PC_1^* \approx 0.102$. Galaxies are assigned by their position relative to these lines:

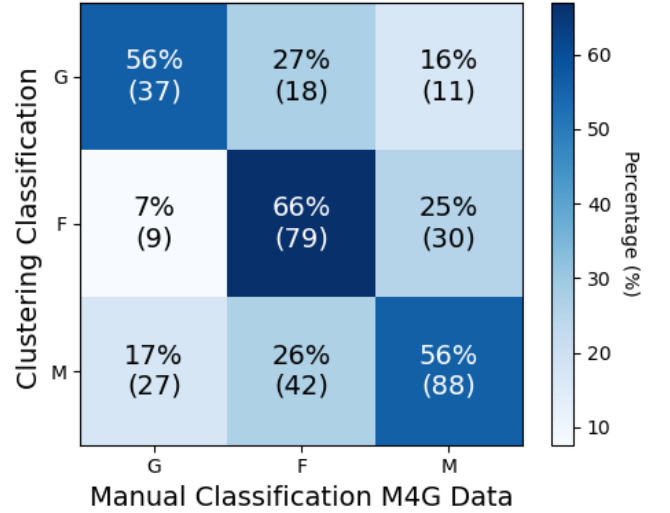


Figure 4. Confusion matrix comparing unsupervised K-means clustering classification with manual morphological classification of spiral galaxies. The matrix shows both percentages and absolute counts (in parentheses) of galaxies.

Table 2. Number of spirals with arm classes identified using our K-means cluster cuts

Arm Class	Number of Galaxies
Grand Design	1129
Flocculent	1668
Multiple Arm	820

- **Grand design galaxies:** $PC_2 > \max\{y_1(PC_1), y_2(PC_1)\}$ (above the upper boundary).
- **Flocculent galaxies:** for $PC_1 < PC_1^*$, $\min\{y_1(PC_1), y_2(PC_1)\} < PC_2 \leq \max\{y_1(PC_1), y_2(PC_1)\}$ (between the two lines; this region vanishes for $PC_1 \geq PC_1^*$).
- **Multiple-arm galaxies:** $PC_2 \leq \min\{y_1(PC_1), y_2(PC_1)\}$ (below both boundaries).

This approach resulted in a total of 1129 grand design galaxies, 1668 flocculent galaxies, and 820 multiple arm galaxies, as summarized in Table 2. However, a subset of galaxies still remained unclassified due to the ambiguity of the k-means clustering in central regions.

4 RESULTS

With our arm classifications in hand, we now explore the correlation between spiral arm classes and properties of both the host galaxy (i.e. baryonic mass fraction, §4.1; gas mass fraction §4.2) and the environment they live in §4.3; i.e. we are asking if a certain type of spiral arm (grand design, multi-arm or flocculent) forms preferentially in galaxies with certain properties, or living in certain environments.

4.1 Dependence of Arm Class on Baryonic Mass Fraction

We first examine the correlation between spiral-arm properties and baryonic mass fraction in disk galaxies. In Fig. 6 we show the fraction

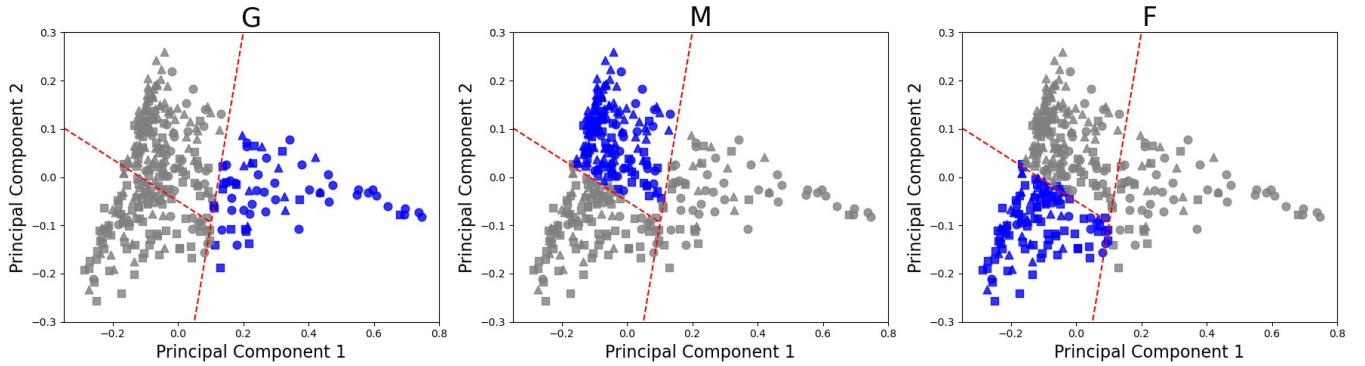


Figure 5. This plot shows the K-means classification results in PCA space for the S4G dataset, which was previously categorized by astronomers (Buta et al. 2015). Symbols denote manual classifications: triangles represent multiple-arm galaxies, squares indicate flocculent galaxies, and circles correspond to grand-design galaxies. The three subplots illustrate the K-means clusters. Blue dots indicate galaxies grouped together by the unsupervised machine learning algorithm, while grey dots represent galaxies assigned to other clusters. This visual highlights the algorithm’s effectiveness in automating galaxy classification within the S4G dataset.

of spiral arms by vote number from GZ:DESI. We find a simple pattern: systems with many arms (more than four, or labelled as “many arms” in GZ:DESI) appear more often at lower baryonic mass fractions, while two- or four-armed spirals appear more often at higher baryonic mass fractions. This suggests that the spiral structure has some connection with the baryonic mass distribution, probably related to formation and secular evolution.

Next, we apply a principal-component (PC) based scheme to separate spirals into three morphological classes. We project the five arm-count fractions onto two PCs and then use two linear decision boundaries,

$$y_1 = 4 PC_1 - 0.5, \quad y_2 = -0.433 PC_1 - 0.05,$$

in the (PC_1, PC_2) plane. As it has been mentioned in 3.1, galaxies are assigned by position relative to the lines:

- **Class M (Grand design):** points above the upper boundary in the local region.
- **Class F (Flocculent):** points between the two lines, only for $PC_1 < PC_1^*$.
- **Class G (Many-armed):** points below both lines.

Overall, the PC approach gives a simple but more objective separation than a single-threshold cut, because it uses the joint information from the five arm-fraction features and draws linear boundaries in the reduced space.

Figure 7 also highlights the distribution of baryonic mass fractions across different arm classes, revealing distinct patterns. Grand design galaxies consistently exhibit higher baryonic mass fractions, while flocculent and multiple arm galaxies show a broader and lower range.

Given the overlap and challenges in separating flocculent and multiple arm galaxies through citizen science votes alone, the use of unsupervised ML methods such as k-means clustering proves invaluable. By refining these classifications, we aim to enhance our understanding of the complex interplay between baryonic mass fractions and spiral arm structures.

Therefore, by focusing on the angular regions within the three clusters, we achieve the separation of grand design, multiple arm, and flocculent galaxies. As illustrated in Figure 7, the distributions and trends for galaxies with different arm numbers regarding the baryonic mass fraction are depicted. Figure 7 reveals a notable pattern: flocculent galaxies are predominantly associated with low baryonic

mass fractions, while grand design galaxies exhibit a tendency towards higher baryonic mass fractions. Multiple arm galaxies fall in between these two categories, aligning with the expected intermediary type between grand design and flocculent galaxy types.

4.2 Dependence of Arm Class on Gas Mass Fraction

Spiral arms may both impact cold gas content, by enhancing star formation (either via concentrating it, or improving efficiency), and high gas content has also been linked to the formation of flocculent spiral types. To investigate the relationship between spiral arm structure and gas content, we studied the distribution of spiral galaxies by arm class across varying gas mass fractions. We analyzed three types of spiral galaxies—grand-design, flocculent, and multiple-arm—by plotting the number of galaxies in each arm class over the total number of galaxies within the same mass fraction bin. Using different colors for each arm class, we created plots showing the trends in galaxy arm classes as a function of baryonic mass, HI mass, stellar mass, and dynamic mass, shows in figure 8.

To ensure our analysis is not biased by galaxy mass differences, we applied a statistical control method. For each mass type (e.g., baryonic, HI, stellar, and dynamic), we controlled for galaxies with identical mass bins. When a mass bin contained an excess number of galaxies of a particular arm class (such as grand-design or flocculent), we randomly excluded a portion of these galaxies, ensuring an equal mass distribution across arm classes within each bin. This process allowed us to isolate the effects of gas mass on arm structure, independent of other variables.

Our findings indicate that the relationship between spiral arm structure and mass varies across mass types:

In the HI and dynamic mass-controlled plots, we observed the same general pattern as in the original dataset without data selection. This consistency suggests that neither HI mass nor dynamic mass significantly influences the distribution of different spiral arm types. In other words, the prevalence of grand-design, flocculent, and multiple-arm galaxies appears independent of variations in HI and dynamic mass, indicating that these mass types are less critical in determining spiral structure. By contrast, the baryonic and stellar mass-controlled plots showed notable differences from the original data. This divergence implies a stronger correlation between baryonic and stellar mass with specific spiral structures. It appears that galax-

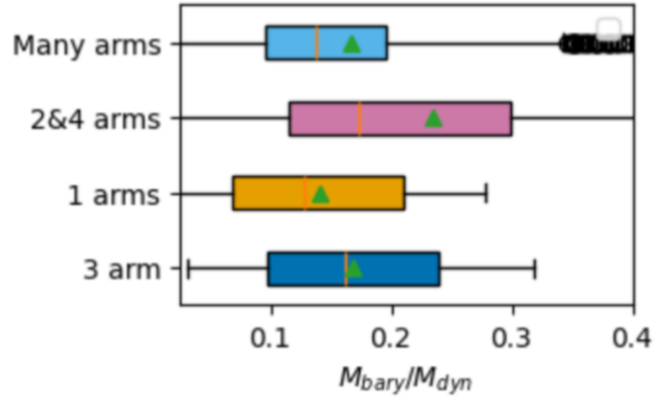
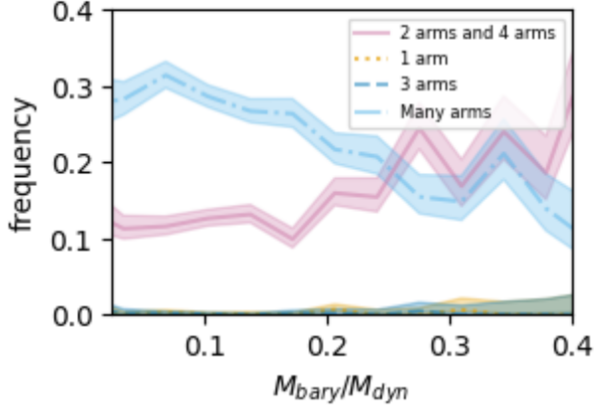


Figure 6. These two plots depict the distribution of spiral galaxies ($p_{\text{spiral}} > 0.5$) with various arm numbers based on the baryonic mass fraction. The left plot shows the fraction of spirals with clearly identified arm numbers (i.e., high vote fraction) as a function of baryonic mass fraction. The right plot presents a box plot illustrating the typical baryonic mass fractions of spiral galaxies across different arm numbers in our sample. The light-colored line in the middle of the box indicates the median value, and the triangle represents the mean value.

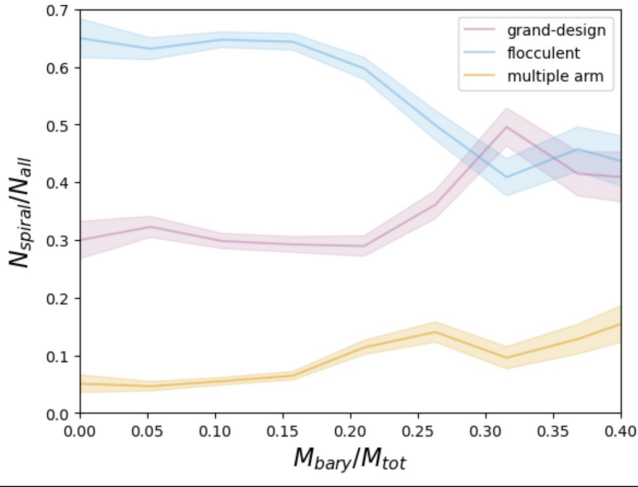


Figure 7. These plots illustrate the relationship between the number frequency of spiral galaxies and their baryonic mass fraction across different arm classes. The orange line corresponds to flocculent galaxies, the pink line represents grand design galaxies, and the blue line indicates multiple arm galaxies.

ies with higher baryonic and stellar mass fractions are more likely to host grand-design or multiple-arm structures, while those with lower fractions tend to exhibit flocculent structures. These results highlight the importance of baryonic mass — particularly its stellar component, which we interpret as the mass of the stellar disc — in shaping spiral arm morphology. While dynamic mass, encompassing both visible and dark matter, shows minimal impact on spiral structure, the visible baryonic components, such as stars and interstellar gas, appear to play a key role in defining the distinct spiral patterns in disk galaxies.

4.3 Dependence of Arm Class on Galaxy Environment

In this analysis, we investigate whether the environmental galaxy number density plays a significant role in determining the spiral structure in disc galaxies, specifically examining three morphological classes: spiral arm, flocculent, and multiple arm galaxies. Figure 9 (refer to the contour plot and density distributions) presents the distribution of the baryonic mass fraction F_{bary} and the normalized baryonic-to-total mass ratio $M_{\text{bary}}/M_{\text{tot}}$ across different environments.

The contour plot highlights the overlap between the three galaxy types, suggesting that the *environmental galaxy number density* does not exhibit significant variations across these morphologies. This observation implies that the local environment, characterized by the density of surrounding galaxies, is not a decisive factor in shaping the presence or absence of spiral arms in disc galaxies. The consistent density distributions shown for all three categories further reinforce this conclusion. Therefore, the influence of environment on the development of spiral structures appears to be marginal.

This finding aligns with prior studies that emphasize intrinsic properties, such as baryonic mass fraction, as more influential in determining the nature of spiral structures in disc galaxies (e.g., ???). It suggests that the dynamics responsible for producing spiral arms are more likely linked to internal processes, including the distribution of baryonic and dark matter, rather than the galaxies' external environments. As demonstrated in the plot, the baryonic mass fraction plays a more central role in differentiating galaxy types, underscoring the significance of internal mass composition in shaping disc morphology.

5 DISCUSSION

In this study, we investigated the relationships between spiral arm morphology and various galaxy properties, specifically baryonic mass fraction, gas mass fraction, and local galaxy environment. Our analysis builds upon existing theoretical frameworks and observational findings regarding spiral structure formation and evolution.

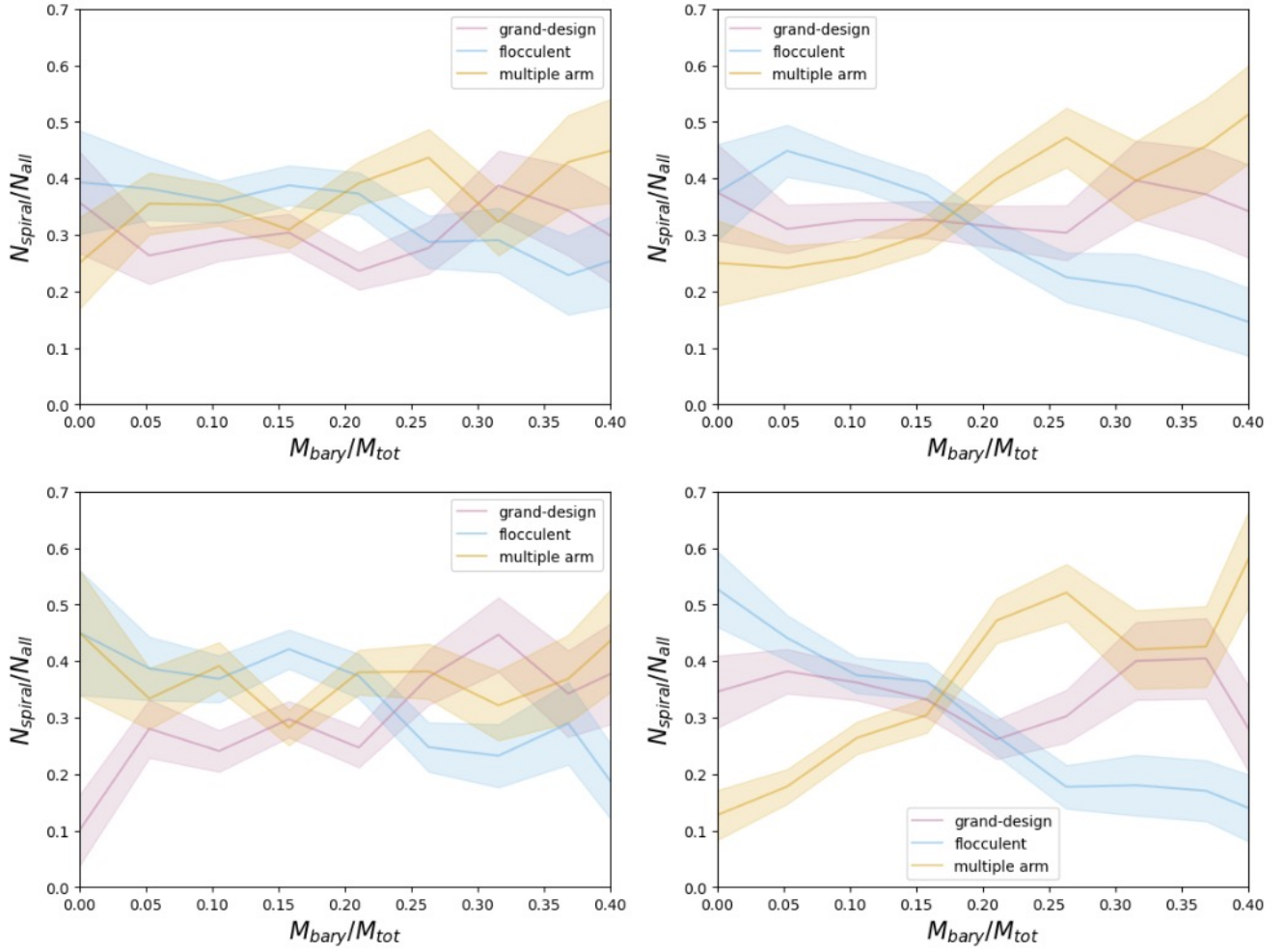


Figure 8. The relationship between the number of spiral galaxies (N_{spiral}) and the total number of galaxies (N_{all}) for different arm classes—grand-design, flocculent, and multiple-arm—plotted as a function of the baryonic mass fraction ($M_{\text{bary}}/M_{\text{tot}}$). Each line represents a specific spiral arm type: red for grand-design, green for flocculent, and blue for multiple-arm galaxies. The four panels depict different control variables used in the analysis: **Top left:** Control for baryonic mass. **Top right:** Control for HI mass. **Bottom left:** Control for stellar mass. **Bottom right:** Control for dynamical mass. By controlling each mass component independently, the plots illustrate the influence of different mass components on the prevalence of distinct spiral arm types in galaxies. Shaded regions around each line represent the uncertainty or variability in the data for each arm class. When we control for both HI mass or dynamical mass we see a pattern consistent with the overall data, while the baryonic and stellar mass controls exhibit different distributions, suggesting a stronger correlation with arm structure.

5.1 Grand Design and Flocculent Galaxies

Our results confirm theoretical predictions regarding grand design and flocculent galaxies. According to density wave theory, grand design galaxies which exhibit well-defined and symmetric spiral arms, result from coherent density waves propagating through the galactic disk (Elmegreen & Elmegreen 1984; Sellwood & Masters 2022). These should be stronger where the disk is heavier relative to the surrounding dark matter halo (CITE). Consistent with this theory, we find that grand design galaxies are more likely to possess higher baryonic mass fractions, likely facilitating the stability and longevity of global density waves. In contrast, flocculent galaxies are characterized by fragmented spiral arms, generally resulting from local gravitational instabilities and patchy star formation regions (Elmegreen & Elmegreen 1984; Dobbs & Baba 2014). These are expected to be more common in galaxies with lighter disks relative to their dark matter halo. Our observations align with these expectations, reveal-

ing a clear tendency for flocculent galaxies to be more common among spirals with lower baryonic mass fractions, likely limiting the development of coherent, global-scale spiral structures.

5.2 Multiple-Arm Galaxies

While flocculent galaxies tend to have lower baryonic mass fractions, and grand design galaxies exhibit a preference for higher baryonic mass fractions; multiple arm galaxies appear to occupy an intermediate position. This may appear in conflict with recent observational findings of Smith et al. (2024), who observe that grand design galaxies tend to have higher central concentrations, larger classical bulges, and tighter spiral arm structures compared to multi-armed systems—characteristics linked to higher baryonic-to-halo mass ratios (Smith et al. 2024). However, the selection of multi-arm galaxies in our and Smith et al. (2024) work differ quite significantly. The

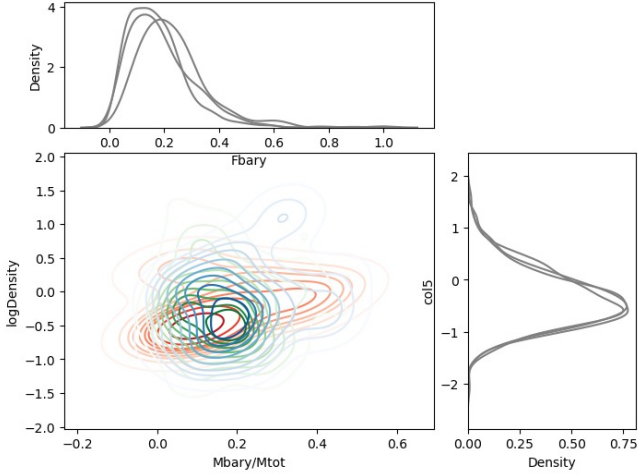


Figure 9. This plot shows that there is no observable relationship between the environment and the arm class. Each arm class is represented with density overlaps, demonstrating the distribution similarity across environmental contexts.

Smith et al. (2024) paper identifies multi-arm galaxies only out of a sample in which Galaxy: 3D (?) volunteers could draw spiral arms. This makes a sample of very clear symmetric and large radius spiral multi-arm galaxies. Our selection in this paper may mix in more flocculent like arms in the multi-arm selection, and explain the different behavior.

Contrary to the conventional morphological classification scheme, where multiple-arm galaxies are positioned as intermediate between grand design and flocculent types (Elmegreen & Elmegreen 1984), our analysis does not show a simple intermediate baryonic mass distribution for these galaxies. Instead, multiple-arm galaxies occupy a broader range, overlapping significantly with both grand design and flocculent classes. This finding suggests a more complex scenario than previously recognized, indicating that multiple-arm galaxies may not represent a straightforward transitional morphology. Rather, multiple-arm galaxies potentially encompass a variety of structural and dynamical conditions not strictly governed by baryonic mass fraction alone. Further investigation employing detailed kinematic and structural analyses could provide additional insights into the underlying mechanisms distinguishing multiple-arm galaxies.

While our current analysis focuses on mass properties and unsupervised classifications, another potentially important factor in shaping spiral arm morphology is the presence and nature of galactic bars. Recent studies have shown that bar dynamics, particularly bar pattern speed and strength, can influence the morphology of spiral arms. For example, slower bars have been associated with tighter spiral arms, possibly due to resonance-driven arm formation mechanisms.

It would be valuable in future work to investigate whether the arm classes identified here—particularly the grand design and multiple-arm categories—correlate with bar strength or bar pattern speed. Such an analysis could reveal whether certain bar configurations preferentially give rise to specific spiral arm types. While this lies beyond the scope of the present study, incorporating bar properties into the PCA or clustering framework may yield deeper insights into the coupled evolution of bars and spiral structure.

5.3 Impact of Galaxy Environment

Our analysis indicates that spiral arm morphology, as quantified by projected galaxy number density, does not show a clear correlation on galaxy environment within our sample. While it is well known that environment significantly affects galaxy morphology at broader classification levels (e.g. Dressler 1980; Lee et al. 2010; Tempel et al. 2011), our results reveal minimal variation in the frequency of spiral arm types across different environmental densities. This suggests that, at least with the environmental metric used in our analysis, local galaxy density does not play a decisive role in determining specific spiral arm classes (grand design, multiple-arm, or flocculent). Instead, our findings reinforce the idea that intrinsic galaxy properties, particularly baryonic mass distributions, dominate the formation and persistence of spiral structures. Further studies using alternative or complementary environmental metrics might be necessary to thoroughly assess the influence of environmental factors on spiral morphology. Early work by Elmegreen & Elmegreen (1987b) found that flocculent spirals preferentially occur in low-density groups, whereas grand design spirals are more common and physically larger in denser environments, underscoring the role of environmental density in shaping spiral structure.

Overall, our study employs a robust methodology integrating citizen science classifications from Galaxy Zoo DESI with unsupervised machine learning techniques such as K-means clustering and PCA to identify types of spiral arms. This approach effectively leverages Galaxy Zoo spiral arm number vote fractions to categorize spiral galaxies into grand design, multiple arm, and flocculent classes. We use these classifications to investigate the role of baryonic mass fractions in shaping spiral structures. We confirm theoretical expectations for grand design and flocculent galaxies being found in higher/lower disk/baryonic mass spirals respectively. We also highlight the complexity of multiple-arm spiral morphologies being sometimes intermediate between grand design and flocculent, but in other selections more distinct. Finally, while our specific measurement of local galaxy environment showed negligible impact on spiral type, further work may be needed to fully explore the role of environment on spiral structure. Never-the-less this finding underscores the importance of intrinsic properties in setting spiral arm types in spiral galaxies.

It's surprising that well over 100 years since spiral structure was first identified, and almost 40 years since the definition of spiral types by (Elmegreen & Elmegreen 1987b) we still don't have a clear observational or theoretical understanding of why a specific galaxy develops spirals of a given type over another.

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